



USER MANUAL

TEAM SYNEX
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DIGITAL COMMUNICATION VISUALIZATION CONSOLE

ABOUT US

Thank you for purchasing the Digital Communication Visualization Console (DCVC). This system is proudly engineered by Synex Engineering.

The DCVC is a standalone bench instrument. It is designed specifically to serve as an interactive desktop laboratory tool. The console translates abstract digital communication theories into clear visual layouts. It displays the entire pipeline, from message character inputs to real-time vector channel transmissions.

Users can easily manipulate data alphabets, runtime parameters, packet frames, and physical analog noise. This hands-on approach allows engineering students, training center operators, and electronics hobbyists to monitor complex bit manipulation in real time.

PACKAGE CONTENT

- 1× Synex Digital Communication Visualization Console (DCVC) Unit
- 1× High-Stability AC-to-DC 12V Power Adapter
- 1× Official Warranty Card
- 1× DCVC Hardware User Manual

SAFETY PRECAUTIONS

Please read these guidelines carefully before operating the console. Following these instructions will protect both the user and the device.

- Only use the included 12V DC power adapter. Using third-party chargers may deliver incorrect voltage and permanently damage the internal circuit boards.
- Do not block the cooling vents on the casing. The internal power regulators require continuous airflow to manage heat efficiently during long laboratory sessions.
- Keep water, coffee, and other liquids away from the console. Spills can cause short circuits on the exposed keypad and internal PCB tracks.
- Always place the device on a stable, flat desktop surface. Do not drop the console.
- Do not attempt to open the enclosure while the device is plugged in.

DEVICE OVERVIEW



FEATURES

- **Real-Time Source Coding:** Supports Fixed-length, Huffman, and Shannon-Fano compression algorithms.
- **Adjustable Data Framing:** Splits data streams into customizable 4-bit, 6-bit, or 8-bit packets.
- **Built-in Error Protection:** Applies Parity, CRC, and Hamming block codes to secure data frames.
- **Multiple Modulations:** Generates real-time waveforms for ASK, BPSK, QPSK, and 16-QAM.
- **Dual Display System:** Utilizes two 3.5" screens to show control menus and signal graphs simultaneously.
- **Live LED Tracking:** Features a 24-LED array to visually monitor binary data streams as they process.
- **Interactive Channel Noise:** Includes a physical analog dial to inject interference and observe real-time signal corruption.

QUICK START GUIDE

Follow these simple steps to safely power on your Digital Communication Visualization Console (DCVC) for the first time.

1.Position the Console:Ensure a stable setup.Place the DCVC unit on a clean, flat laboratory desk. Ensure the cooling vents on the casing are not blocked.

2.Connect the Power:Use only the provided 12V adapter.Plug the included power adapter into a standard wall outlet. Then, insert the connector into the DC power port on the back of the console.

3.Turn on the Device:Press the power switch located on the back panel next to the power port.

4.Observe the Boot Sequence:Hardware initialization.The 24 LED indicators will flash briefly as the system runs a hardware check. Both 3.5" TFT screens will light up and display the Synex Engineering logo.

5.Ready for Input:After a few seconds, the left screen will load the Main Menu. The right screen will show a standby grid. The device is now ready, and you can use the keypad to start your first simulation.

SOURCE CODING

The console uses a specific alphabet set for data input: $S = \{A, B, C, D, E\}$. Follow these steps to compress your data.

1. Open the Menu: Navigate to the Source Compiling Menu on the left screen (Screen 1).
2. Choose an Algorithm: Fixed-Length, Huffman, or Shannon-Fano. Select your desired compression algorithm using the keypad.
3. Set Probabilities: Required for Huffman and Shannon-Fano. If prompted, type in the probability for each character. Ensure all the probabilities add up to exactly 1.0.
4. Type Your Message: Use the keys [A] through [E] to type your custom message sequence.
5. Process the Data: Press the [#] key. The system will process your message, and the generated binary compression tree will appear instantly on Screen 1.

FRAMING METHODS

After compressing your data, the system allows you to split the data stream into distinct packets. This feature operates in real time. Select one of the following packet sizes from the menu to set your frame boundaries.

- 4-bit Packets: Best for observing short-block transmissions and basic data tracking.
- 6-bit Packets: An intermediate layout for testing medium-sized data frames.
- 8-bit Packets: The traditional, full-byte formatting configuration.

CHANNEL CODING

To protect your data packets from errors, you must add mathematical protection bits.

1. Open Channel Coding: Navigate to the Channel Coding Sub-Menu on Screen 1.

2. Select a Protection Mode: Parity or CRC. Choose either standard Parity Validation (Even/Odd) or a Cyclic Redundancy Check (CRC-3 or CRC-4).

3. Enter the Polynomial: For CRC modes only. If you selected CRC, use the keypad to type your generator polynomial code. For example, typing '1011' will set up a 4-bit CRC generator.

4. Apply the Protection: Press the [#] key. The system will automatically calculate the remainder codes and attach them to your packets.

HARDWARE SPECIFICATIONS

The following details the core electrical and mechanical specifications of the DCVC unit.

- Core Controller Engine: STM32 ARM Cortex-M Series (32-bit Architecture)
- Input Operating Voltage: 12V DC Sourced Line (via the enclosed adapter)
- Display Panel Hardware: Dual 3.5” SPI TFT Screens (320 × 480 resolution)
- Bit Tracking Diagnostic: 24-Channel LED Matrix driven via three cascaded 74HC595 ICs

MAINTENANCE & TROUBLESHOOTING

Follow these guidelines to keep your console operating smoothly and to resolve common issues.

- **Cleaning the Enclosure:** Wipe the 3D-printed chassis with a soft, dry microfiber cloth. Do not use chemical solvents or abrasive sprays, especially near the glass screens.
- **Powering Down:** Unplug the 12V power adapter from the wall outlet if the console will not be used for an extended period.

WARRANTY & SUPPORT

Legal Exclusions: Synex Engineering is not responsible for hardware damage caused by incorrect power supplies, liquid exposure, or unauthorized modifications. Furthermore, we are not liable for academic laboratory failures or grading issues related to the use of this device.

Limited Warranty Information: The DCVC unit comes with a 2-year limited warranty from the purchase date. This warranty covers internal manufacturing defects and component failures that occur under normal laboratory use.

The warranty will be immediately voided if:

- The 3D-printed enclosure is opened or modified.
- The device suffers physical impact damage (e.g., dropping).
- A third-party or incorrect power adapter is used.

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